ProLecture

A benchmark of proactivity that runs into reasoning

The model watches a lecture in real time and decides for itself **when** to answer and **what** to answer.

The material is 3Blue1Brown lectures. Status: 3 videos through the whole cycle, 44 samples, the annotation pipeline assembled.

The niche

proactivity where the answer requires a derivation rather than a reading

Work	What it measures	What is missing
OVO-Bench	backward / real-time / forward	MCQ, no audio, perception
ProactiveVideoQA	open-ended, the PAUC metric	simple questions “tell me when X”
OmniPro	6 abilities $\times$ 9 subtasks	84% audio, yet vision-only keeps up

- A lecture has almost **no perceptual difficulty**: clean rendering, large formulas, clear speech.
- What it does have is a logical structure: premise $\rightarrow$ derivation $\rightarrow$ consequence.
- So the only way to answer in time is to **understand the content** — not to recognise an object.

Why 3Blue1Brown

a unique asset: we have the source code of the picture

For **140** videos both the YouTube link and an existing path to the manim source of the scene are known.

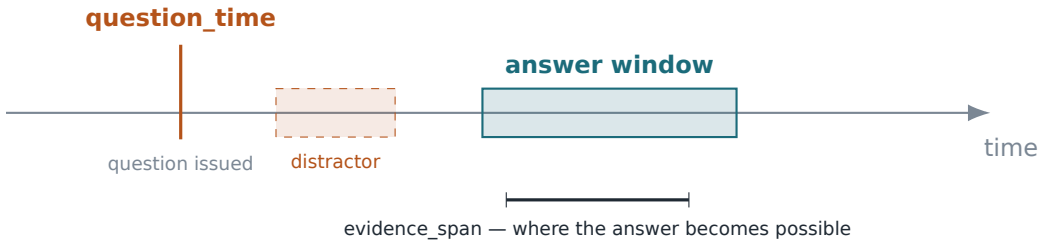
None of the existing benchmarks has the sources of the video track.

- an exact ground truth of what is on the screen (the Tex strings with LaTeX);
- semantic boundaries: scene names, the order of the animations;
- **a re-render with an injected error** — content guaranteed not to have been in pre-training.

The connection “lesson ↔ video ↔ code” exists only in the frontmatter of the site: the `3b1b/videos` repository itself contains not a single link to YouTube.

The anatomy of a sample

the question in one second, the answer in a window, the evidence separately



- The `temporal_rel` axis (`pre` / `at` / `post`) is **checked by the validator**: it recomputes it from `question_time` and `evidence_span` and rejects a mismatch.
- `distractor_windows` — moments where an answer looks apt but is premature.
- The confirmation of the reference answer lives separately, in `provenance`: it may lie later than the window and even outside the video.

What else is in a sample

what the answer is judged by, and what is measured rather than declared

In every window

`response` — the reference answer
`rubric` — what the judge scores against
`trigger_modality` — visual / speech / sound
`depends_on` — a link to an earlier window

The rubric replaces the MCQ. From `q01`:

> “the number 2.5 is named”
> “if a computation is shown, it is weighted, with 1/2, 1/4, 1/8, 1/8”
> **“do NOT credit 1.75 — that is the original code, not the swapped one”**

`validation` — the measurements

One flag per probe, each filled in by a model’s answer, not by an opinion:

`solvable` — solvable at all
`blind_solved` — without the video
`textonly_solved` — the transcript is enough
`visiononly_solved` — the frames are enough
`frozenframe_solved` — one frame is enough

`audio_dependency` is derived from these; the annotator’s opinion lives in `audio_dependency_claim` and influences no conclusion.

`status`: `draft` → `validated` / `rejected`.

The three taxonomy axes

every sample is a point in the cube (act, relation, level of reasoning)

A. The type of act

`alert` — an alert
`monitor` — a state
`ground` — grounding
`narrate` — narration
`predict` — prediction
`verify` — verification
`counterfact` — a counterfactual

B. The relation to the trigger

`pre` — the evidence comes after the question: **the ability to wait**

`at` — here and now

`post` — an answer from memory

One question in two regimes gives a controlled pair.

C. The level of reasoning

`L1` perception — read off

`L2` integration — connect two moments

`L3` derivation — carry out reasoning that is not in the video

The criterion for L3: the blind baseline does not solve it.

Axis C is what OmniPro does not have: it is precisely along it that its models fail to differ.

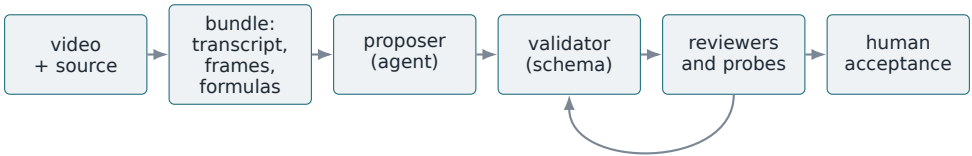
The principles the design grew out of

each point closes a failure that sank one of the works reviewed

- **Labels are set by measurement, not by declaration.** `audio_dependency` is derived from the modality probes; the annotator's opinion lives in a separate field and influences no conclusion.
- **Generator $\neq$ judge $\neq$ participant** — a rule for the measurement phase. At present the proposer, the reviewers and the coordinator are one model, and no judge or participant has been run yet. The judge may be a pure text LLM: it reads the rubric and the answer, not the video.
- **Open answers with a rubric, no MCQs.** A skewed answer key breaks a benchmark quietly.
- **Mandatory dummy baselines:** blind, text-only, vision-only, frozen-frame, "answer immediately", "answer at the end", "stay silent".
- **No quotas** of the form "exactly N samples per task": however many passed validation is however many there are.
- **Contamination is measured, not denied:** a 2026 pilot video and a 2017 control one, so the blind-baseline gap becomes a number.

The pipeline

from a video to a sample that has been through review



- **The validator** plays the role of tests: non-overlapping windows, references only backwards, a distractor not coinciding with a trigger, `temporal_rel` agreeing with the evidence. It returns code 1 on any error.
- **The reviewers and the probes** are covered shortly: three roles check the annotation from different sides, four probes fill in the `validation` flags by measurement.
- **The round repeats** until every verdict is `PASS`; a repeat round goes to fresh subagents that have not seen the previous discussions.

How this works on live samples

one and the same blind check, opposite outcomes

q01 — passed the check

“As soon as a table of code words of different lengths with their probabilities first appears, compute what the average number of bits becomes if ‘up’ and ‘left’ are swapped.”

Question at 250 s, table at 314 s; the answer of 2.5 bits is not in the video.

The semi-blind reviewer named 2.5 — and was not credited: the answer opened with “I cannot name an exact number”, then a conditional guess and a branch with 2.0. **The problem was guessed, the answer was not found.**

q08 — rejected

“Say before the author tells it himself: what source of letter probabilities did Shannon turn to, and why does he not have the frequency-table problem?”

Before any viewing the reviewer produced **both** halves of the answer: a live native speaker, and the growth of the number of contexts as 27^n. Three rubric items out of three, without a single assumption about a frame.

`blind_solved = true`, `status: rejected`; the level was lowered `L3` → `L1` along the way.

The rule: leaning on a commonly known fact is contamination; guessing the problem is not.

Three reviewers, blind to each other

they run in parallel; none of them edits the files

- **The semi-blind one.** Sees the material up to `t_end` but does not know `t_start` — and names for themselves the moment the answer became possible. A checker who knows the answer "recognises" it on a half-drawn frame and keeps moving the left boundary of the window leftwards; this one does not have that knowledge.

  They separately record `prior_answer` — their own answer **before** viewing, verbatim. It is what the coordinator derives `blind_solved` from.
- **The sample reviewer.** Everything relating to the question as a whole: the wording, the rubric, the decoys, the reference answer, the links between windows. The window boundaries are not their work.
- **The per-window reviewer, one per window.** The role was separated not for symmetry: with a single reviewer on a question with nine windows, the frame budget was enough for the first, and after that they made things up.

Each returns a structured review against the contract (`schema/review.py`); the coordinator applies the edits, so it is visible who found what.

Four probes: the label is measured

each is given exactly one channel, and we see whether it answers

Probe	Material	Frames	Fills in
`text`	transcript up to `t_end`	0	`textonly_solved`
`frame`	one frame mid-window	1	`frozenframe_solved`
`vision`	frames up to `t_end`	up to 40	`visiononly_solved`
`full`	transcript and frames	up to 40	`solvable`

`solvable = textonly OR visiononly OR` the `full` answer; `solvable = false` is a `CRITICAL_FAIL` — the question is not solvable even from the full material.

`visiononly_solved` together with `solvable` give **`audio_dependency`** — this label is never set by hand.

`frozenframe_solved = true` at level `L2`/`L3` means the level is inflated.

A probe's isolation rests on three things: the prompt asks it not to reach into another channel, the script does not let two modes share a folder, and the probe must confess in `violations`.

Tools that did not exist

two places where what was available did not fit

Our own scene change detector

Off-the-shelf detectors compare a frame with the previous one, while manim's typical transition is a smooth 2–4 second travel where neighbouring frames barely differ.

We compute the **accumulated drift** from the last captured frame and capture once the animation has settled.

One threshold per domain: **13.5** / **12.9** / **10.1** frames per minute on the videos from 2026 and 2017.

On the control stretch: **18** states out of 18 against **10** for the uniform grid.

A dense description of the video track

Like subtitles, only with what happens on the screen instead of utterances: the annotator reads text instead of looking through dozens of frames.

Two frames per second, overlapping 5-second windows, strict JSON against a schema (guided decoding).

The pilot: **3878** frames, **485** fragments, **20** minutes on two A100s, **1939** seconds out of 1939 covered by events.

The describing model is not shown the subtitles, so the description can stand in for the frames in the vision-only probe.

Where we are now

Done

the pipeline: mapping $\rightarrow$ download $\rightarrow$ bundle;

the schema and the semantic validator;

prompts: proposer, three reviewers, four probes;

the scene detector, the video-track description;

3 videos through the whole cycle (entropy $\times 2$, 2026; Taylor series, 2017): **44** samples, **43** validated, **61** triggers, **42** distractors;

first runs of streaming video models: **the current tasks are too hard for them**.

Next

run the dummy baselines and see that they fail;

measure the blind gap on the 2017 control video;

the streaming harness and an LLM judge against the rubric;

expanding the annotation: the share of L3 must grow;

re-rendering scenes with an injected error.

Open assumptions go to Limitations rather than being swept away: the leak of the distractor's explanation, the difficulty (now confirmed on streaming models), the small volume.